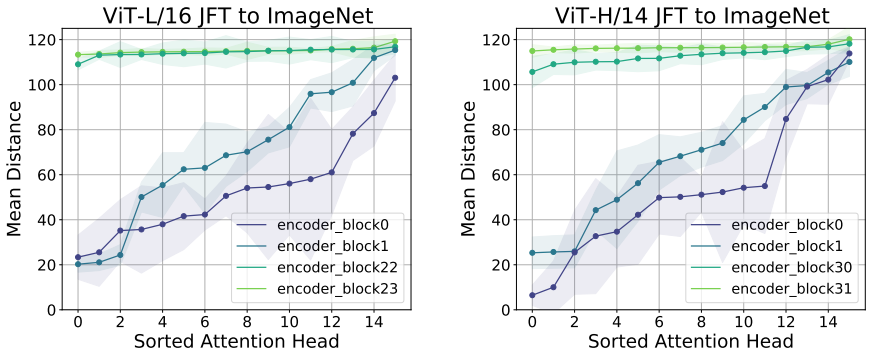

Figure 3: Plotting attention head mean distances shows lower ViT layers attend both locally and globally, while higher layers primarily incorporate global information. For each attention head, we compute the pixel distance it attends to, weighted by the attention weights, and then average over 5000 datapoints to get an average attention head distance. We plot the heads sorted by their average attention distance for the two lowest and two highest layers in the ViT, observing that the lower layers attend both locally and globally, while the higher layers attend entirely globally.

all ResNet layers, likely because this set of layers mainly manipulates the CLS token representation, further studied in Section 6.

Taken together, these results suggest that (i) ViT lower layers compute representations in a different way to lower layers in the ResNet, (ii) ViT also more strongly propagates representations between lower and higher layers (iii) the highest layers of ViT have quite different representations to ResNet.

5 Local and Global Information in Layer Representations

In the previous section, we observed much greater similarity between lower and higher layers in ViT, and we also saw that ResNet required more lower layers to compute similar representations to a smaller set of ViT lower layers. In this section, we explore one possible reason for this difference: the difference in the ability to incorporate global information between the two models. How much global information is aggregated by early self-attention layers in ViT? Are there noticeable resulting differences to the features of CNNs, which have fixed, local receptive fields in early layers? In studying these questions, we demonstrate the influence of global representations and a surprising connection between scale and self-attention distances.

Analyzing Attention Distances: We start by analyzing ViT self-attention layers, which are the mechanism for ViT to aggregate information from other spatial locations, and structurally very different to the fixed receptive field sizes of CNNs. Each self-attention layer comprises multiple self-attention heads, and for each head we can compute the average distance between the query patch position and the locations it attends to. This reveals how much local vs global information each self-attention layer is aggregating for the representation. Specifically, we weight the pixel distances by the attention weights for each attention head and average over 5000 datapoints, with results shown in Figure 3. In agreement with Dosovitskiy et al. [14], we observe that even in the lowest layers of ViT, self-attention layers have a mix of local heads (small distances) and global heads (large distances). This is in contrast to CNNs, which are hardcoded to attend only locally in the lower layers. At higher layers, all self-attention heads are global.

Interestingly, we see a clear effect of scale on attention. In Figure 4, we look at attention distances when training *only* on ImageNet (no large-scale pre-training), which leads to much lower performance in ViT-L/16 and ViT-H/14 [14]. Comparing to Figure 3, we see that with not enough data, ViT *does not learn to attend locally* in earlier layers. Together, this suggests that using local information early on for image tasks (which is hardcoded into CNN architectures) is important for strong performance.

Does access to global information result in different features? The results of Figure 3 demonstrate that ViTs have access to more global information than CNNs in their lower layers. But does this result in different learned features? As an interventional test, we take subsets of the ViT attention heads from the first encoder block, ranging from the subset corresponding to the most local attention

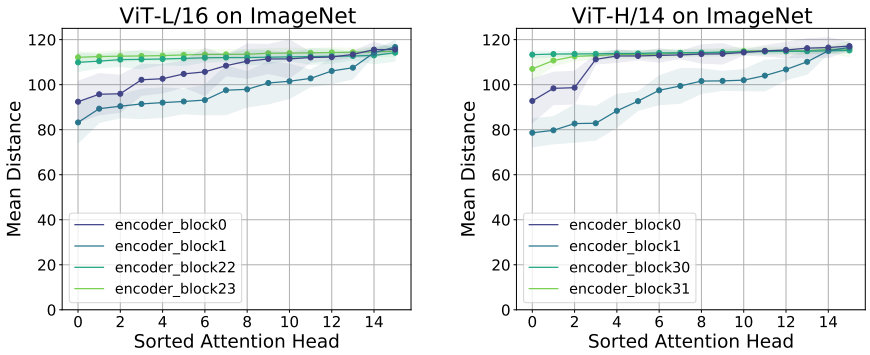

Figure 4: With less training data, lower attention layers do not learn to attend locally. Comparing the results to Figure 3, we see that training only on ImageNet leads to the lower layers not learning to attend more locally. These models also perform much worse when only trained on ImageNet, suggesting that incorporating local features (which is hardcoded into CNNs) may be important for strong performance. (See also Figure C.5.)

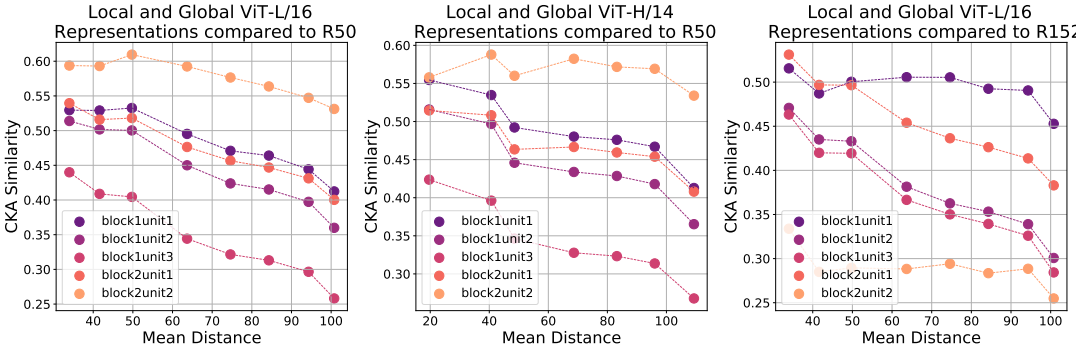

Figure 5: Lower layer representations of ResNet are most similar to representations corresponding to local attention heads of ViT. We take subsets of ViT attention heads in the first encoder block, ranging from the most locally attending heads (smallest mean distance) to the most global heads (largest mean distance). We then compute CKA similarity between these subsets and lower layer representations in the ResNet. We observe that lower ResNet layers are most similar to the features learned by local attention heads of ViT, and decrease monotonically in similarity as more global information is incorporated, demonstrating that the global heads learn quantitatively different features.

heads to a subset of the representation corresponding to the most global attention heads. We then compute CKA similarity between these subsets and the lower layer representations of ResNet.

The results, shown in Figure 5, which plot the mean distance for each subset against CKA similarity, clearly show a monotonic decrease in similarity as mean attention distance grows, demonstrating that access to more global information also leads to quantitatively different features than computed by the local receptive fields in the lower layers of the ResNet.

Effective Receptive Fields: We conclude by computing *effective receptive fields* [23] for both ResNets and ViTs, with results in Figure 6 and Appendix C. We observe that lower layer effective receptive fields for ViT are indeed larger than in ResNets, and while ResNet effective receptive fields grow gradually, ViT receptive fields become much more global midway through the network. ViT receptive fields also show strong dependence on their center patch due to their strong residual connections, studied in the next section. As we show in Appendix C, in attention sublayers, receptive fields taken before the residual connection show far less dependence on this central patch.

6 Representation Propagation through Skip Connections

The results of the previous section demonstrate that ViTs learn different representations to ResNets in lower layers due to access to global information, which explains some of the differences in represen-

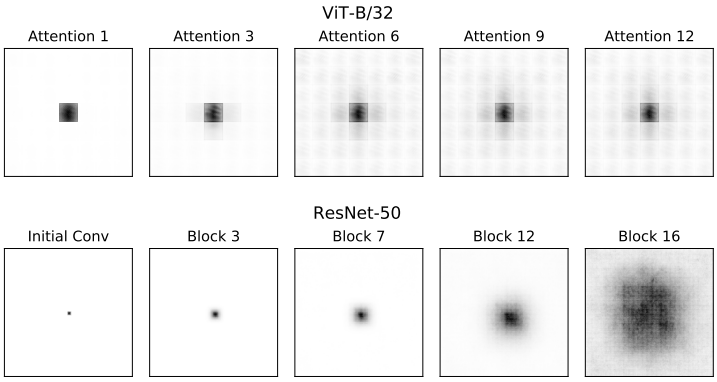

Figure 6: ResNet effective receptive fields are highly local and grow gradually; ViT effective receptive fields shift from local to global. We measure the effective receptive field of different layers as the absolute value of the gradient of the center location of the feature map (taken after residual connections) with respect to the input. Results are averaged across all channels in each map for 32 randomly-selected images.

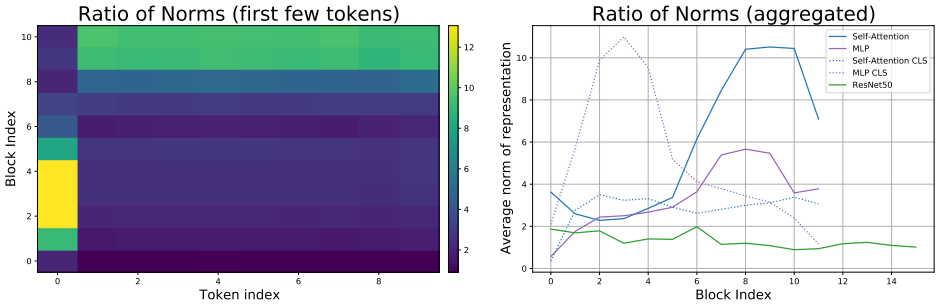

Figure 7: Most information in ViT passes through skip connections. Comparison of representation norms between the skip-connection (identity) and the long branch for ViT-B/16 trained on ImageNet and a ResNet. For ViT, we show the CLS token separately from the rest of the representation. (left) shows the ratios separated for the first few tokens (token 0 is CLS), (right) shows averages over all tokens.

tation structure observed in Section 4. However, the highly uniform nature of ViT representations (Figure 1) also suggests lower representations are faithfully propagated to higher layers. But how does this happen? In this section, we explore the role of skip connections in representation propagation across ViTs and ResNets, discovering ViT skip connections are highly influential, with a clear phase transition from preserving the CLS (class) token representation (in lower layers) to spatial token representations (in higher layers).

Like Transformers, ViTs contain *skip* (aka *identity* or *shortcut*) connections throughout, which are added on after the (i) self-attention layer, and (ii) MLP layer. To study their effect, we plot the norm ratio $||z_i||/||f(z_i)||$ where z_i is the hidden representation of the ith layer coming from the skip connection, and $f(z_i)$ is the transformation of z_i from the *long branch* (i.e. MLP or self-attention.)

The results are in Figure 7 (with additional cosine similarity analysis in Figure E.2.) The heatmap on the left shows $||z_i||/||f(z_i)||$ for different token representations. We observe a striking phase transition: in the first half of the network, the CLS token (token 0) representation is primarily propagated by the skip connection branch (high norm ratio), while the spatial token representations have a large contribution coming from the long branch (lower norm ratio). Strikingly, in the second half of the network, this is reversed.

The right pane, which has line plots of these norm ratios across ResNet50, the ViT CLS token and the ViT spatial tokens additionally demonstrates that skip connection is much more influential in ViT compared to ResNet: we observe much higher norm ratios for ViT throughout, along with the phase transition from CLS to spatial token propagation (shown for the MLP and self-attention layers.)

ViT Representation Structure without Skip Connections: The norm ratio results strongly suggest that skip connections play a key role in the representational structure of ViT. To test this interventionally, we train ViT models with skip connections removed in block i for varying i, and plot the CKA representation heatmap. The results, in Figure 8, illustrate that removing the skip connections in a block partitions the layer representations on either side. (We note a performance drop of 4% when